

Practical - 7

1. Memory Address

```
harshika@siri:~$ gedit memory_segments.c
** (gedit:519): WARNING **: 03:52:39.694: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:519): WARNING **: 03:52:39.694: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$ gcc memory_segment.c -o memory_segment
ccl: fatal error: memory_segment.c: No such file or directory
compilation terminated.
harshika@siri:~$ gcc memory_segments.c -o memory_segments
harshika@siri:~$ ./memory_segments
===== Linux Process Memory Addresses =====
Address of code (function): 0x5cfff4903f209
Address of global variable: 0x5cfff49442010
Address of static global variable: 0x5cfff49442014
Address of static local variable: 0x5cfff49442018
Address of heap variable: 0x5cfff82dbf010
Address of stack variable: 0x7ffdf42e66ec
Process ID (PID): 773
Keep this process running to inspect /proc/773/maps.
Press Enter to exit...
harshika@siri:~$ gedit memory_segments.c
** (gedit:785): WARNING **: 03:54:08.670: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:785): WARNING **: 03:54:08.670: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
```

2. Address Space Layout

```
harshika@siri:~$ cat /proc/1357/maps
5af02cab000-5af02cabd000 r--p 00000000 08:30 26066 /home/harshika/memory_segments
5af02cabd000-5af02cabe000 r-xp 00001000 08:30 26066 /home/harshika/memory_segments
5af02cabe000-5af02cabf000 r--p 00002000 08:30 26066 /home/harshika/memory_segments
5af02cabf000-5af02cac0000 r-p 00002000 08:30 26066 /home/harshika/memory_segments
5af02cac0000-5af02cac1000 rw-p 00003000 08:30 26066 /home/harshika/memory_segments
5af05c6cf000-5af05c6cf0000 rw-p 00000000 00:00 0 [heap]
7d958fa00000-7d958fa28000 r--p 00000000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7d958fa28000-7d958fbc0000 r-xp 00028000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7d958fbc0000-7d958fc0e000 r--p 001c0000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7d958fc0e000-7d958fc12000 r--p 0020d000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7d958fc12000-7d958fc14000 rw-p 00211000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7d958fc14000-7d958fc21000 rw-p 00000000 00:00 0
7d958fc21000-7d958fc75000 rw-p 00000000 00:00 0
7d958fc75000-7d958fc7e000 rw-p 00000000 00:00 0
7d958fc7e000-7d958fc82000 r--p 00000000 00:00 0 [vvar]
7d958fc82000-7d958fc84000 r--p 00000000 00:00 0 [vvar_vclock]
7d958fc84000-7d958fc86000 r-xp 00000000 00:00 0 [vdso]
7d958fc86000-7d958fc87000 r--p 00000000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7d958fc87000-7d958fcb6000 r-xp 00001000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7d958fcb6000-7d958fcc1000 r--p 0003b000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7d958fcc1000-7d958fcc3000 r--p 0003b000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7d958fcc3000-7d958fcc4000 rw-p 0003d000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7d958fcc4000-7d958fcc5000 rw-p 00000000 00:00 0
7ffdf0d151000-7ffdf0d172000 rw-p 00000000 00:00 0 [stack]
harshika@siri:~$
```